Advanced Game Theory

Solution concepts

Lecture 2

Minimax

Rationalizability

Bayesian Nash equilibrium

Correlated equilibrium

Olivier Gossner

From information to solution concepts

Lecture 1

Partitions
Knowledge and common knowledge
Conditional beliefs

Lecture 2

Predictions of play
Beliefs and equilibrium
Role of information

How does information enter these solution concepts?

Four lectures

1 **Knowledge and agreement**
Partitions; common knowledge; Aumann's theorem

2 **Solution concepts**
Minimax; rationalizability; Bayesian games; correlated equilibrium

3 **Comparing and choosing information**
Blackwell; strategic value; information design

4 **Higher-order beliefs and universal representation**
Email game; universal type space; interim rationalizability

1 **Minimax**
Guarantees; mixing; a separation argument

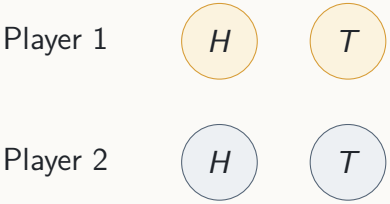
2 **Rationalizability**
Rational choice; iterated deletion; largest fixed point

3 **Bayesian games**
Types; contingent plans; best responses at each type

4 **Correlated equilibrium**
Signals without payoff news; obedience; revelation

Matching pennies

Both players choose heads (H) or tails (T) simultaneously.

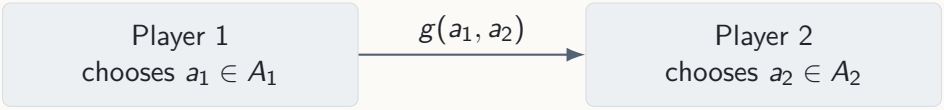


	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

Player 1: row. Player 2: column.

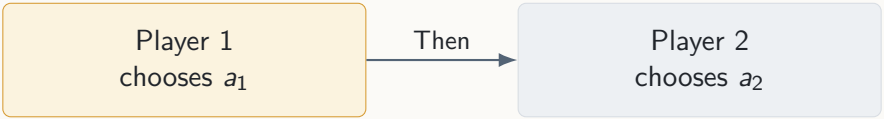
Player 1 wins when the faces match; Player 2 wins when they differ.

A zero-sum game



How much can Player 1 guarantee? How much can Player 1 defend?

Guarantee: Player 1 chooses first

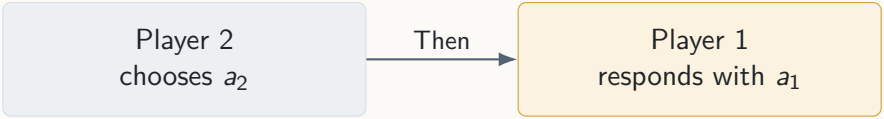


Player 1 can guarantee w if

$$\exists a_1 \quad \forall a_2 : \quad g(a_1, a_2) \geq w.$$

Best pure-strategy guarantee: $\underline{v}^P = \max_{a_1 \in A_1} \min_{a_2 \in A_2} g(a_1, a_2)$.

Defend: Player 1 responds to Player 2



Player 1 can defend w if

$$\forall a_2 \quad \exists a_1 : \quad g(a_1, a_2) \geq w.$$

Best pure-strategy defense: $\overline{v}^P = \min_{a_2 \in A_2} \max_{a_1 \in A_1} g(a_1, a_2)$.

Matching pennies: a gap with pure strategies

	H	T
H	1	-1
T	-1	1

Player 1's payoffs.

Guarantee

Player 2 chooses the other face.

$$\underline{v}^P = -1.$$

Defend

Player 1 matches Player 2's face.

$$\overline{v}^P = 1.$$

It is always easier for Player 1 to defend

Guarantee

$$\exists a_1 \; \forall a_2 : \; g(a_1, a_2) \geq w.$$

Defend

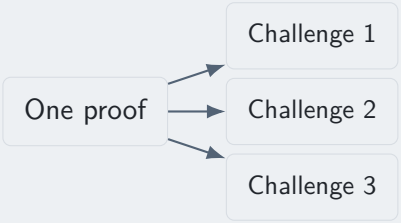
$$\forall a_2 \; \exists a_1 : \; g(a_1, a_2) \geq w.$$

A guaranteeing strategy works against every action of Player 2.

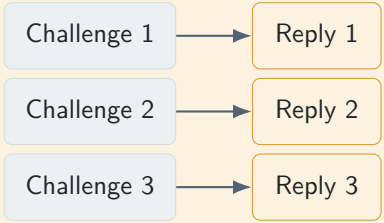
$$\underline{v}^P \leq \overline{v}^P$$

Proofs and counterexamples: a quantifier-order analogy

Guarantee: find a proof

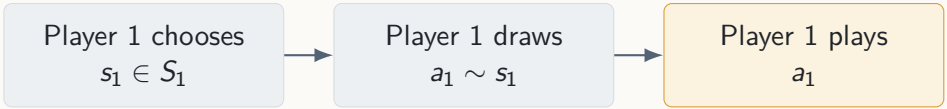


Defend: beat counterexamples



Mixing: Player 1 chooses a distribution

$$S_i = \Delta(A_i), \qquad g(s_1, s_2) = \sum_{a_1, a_2} s_1(a_1) s_2(a_2) g(a_1, a_2).$$



To defend, Player 1 responds to Player 2's distribution before Player 2 draws an action.

Guarantee and defend with mixed strategies

Player 1's guarantee

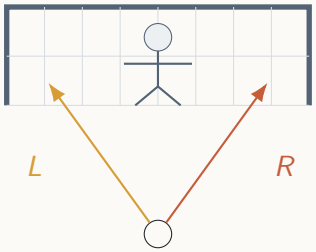
$$\underline{v} = \max_{s_1} \min_{s_2} g(s_1, s_2).$$

Player 1's defense

$$\overline{v} = \min_{s_2} \max_{s_1} g(s_1, s_2).$$

$\underline{v} \leq \overline{v}$ Does mixing close the gap?

Penalty kicks: 80% versus 60%

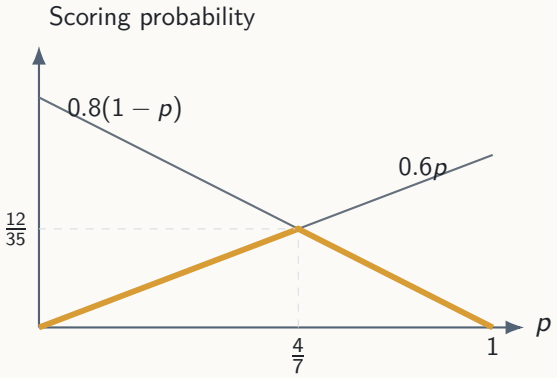


	l	r
L	0%	80%
R	60%	0%

Player 1's scoring probability

Player 1 (shooter): row. Player 2 (goalkeeper): column.

Player 1's guarantee



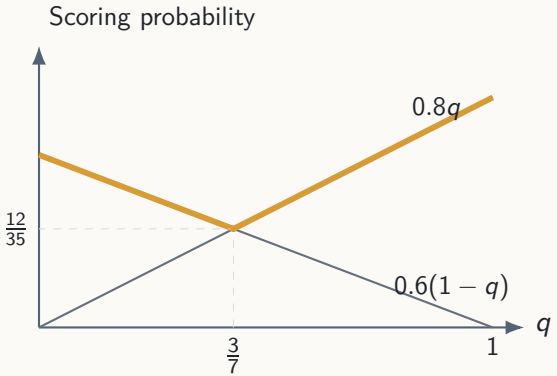
$p = \Pr(\text{Player 1 plays } R)$.

Against l: $0.6p$.

Against r: $0.8(1-p)$.

Player 1 uses $p = \frac{4}{7}$ and guarantees at least $\frac{12}{35} \simeq 34.3\%$.

Player 1's defense



$q = \Pr(\text{Player 2 plays } r)$.

Given Player 2's mixture q,
Player 1 chooses the better kick:

$$\max\{0.8q,\ 0.6(1-q)\}.$$

The worst q is $\frac{3}{7}$.

Against every mixture of Player 2, Player 1 can score at least $\frac{12}{35} \simeq 34.3\%$.

The gap closes

Player 1's optimal mixture

$$p^* = \Pr(R) = \frac{4}{7}.$$

Guarantees at least $\frac{12}{35}$ against Player 2.

Player 2's optimal mixture

$$q^* = \Pr(r) = \frac{3}{7}.$$

Limits Player 1's payoff to $\frac{12}{35}$.

Player 1's guarantee = Player 1's defense: $\underline{v} = \overline{v} = \frac{12}{35}$.

Why does Player 1 use the less effective kick more often?

The minimax theorem

von Neumann (1928) Finite action sets; mixed strategies.

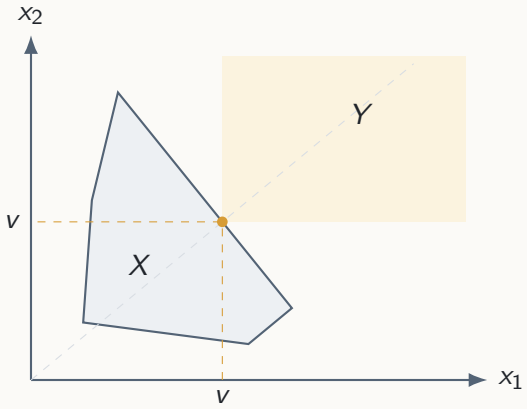
$$\underbrace{\max_{s_1 \in S_1} \min_{s_2 \in S_2} g(s_1, s_2)}_{\text{guarantee}} = \underbrace{\min_{s_2 \in S_2} \max_{s_1 \in S_1} g(s_1, s_2)}_{\text{defend}} = v.$$

Optimal strategies s_1^*, s_2^* satisfy

$$g(s_1, s_2^*) \le v \le g(s_1^*, s_2) \qquad \text{for every } s_1, s_2.$$

(s_1^*, s_2^*) is a Nash equilibrium.

Proof: payoff vectors and an unattainable region



One coordinate per action $j \in A_2$ of Player 2:

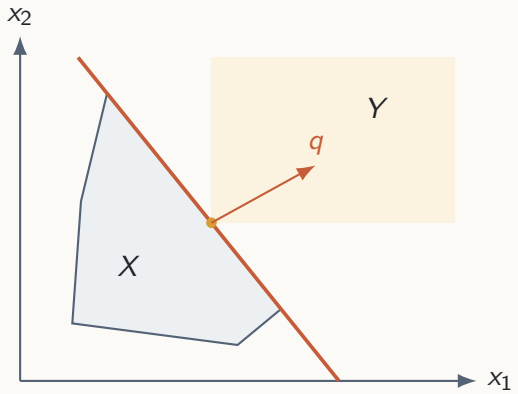
$$X = \{(g(s_1, j))_j : s_1 \in S_1\}.$$

$$v = \max_{x \in X} \min_j x_j.$$

$$Y = \{y : y_j > v \text{ for all } j\}.$$

X is compact and convex; $X \cap Y = \varnothing$.

Proof: the separator is Player 2's mixture



Separation gives $q \neq 0$ with

$$q \cdot x \leq q \cdot y \quad (x \in X,\ y \in Y).$$

Let each coordinate of y grow: $q_j \geq 0$.

Normalize: $\sum_j q_j = 1$.

Let $y \downarrow (v, \ldots, v)$:

$$g(s_1, q) = q \cdot x \leq v.$$

$\overline{v} \leq v = \underline{v}$ and hence equality.

1 **Minimax**
Guarantees; mixing; a separation argument

2 **Rationalizability**
Rational choice; iterated deletion; largest fixed point

3 **Bayesian games**
Types; contingent plans; best responses at each type

4 **Correlated equilibrium**
Signals without payoff news; obedience; revelation

I. What does rational mean?

A finite game, with payoffs $g_i(a_i, a_{-i})$.

Undominated

No mixed strategy strictly dominates the chosen action.

$\overset{?}{\Longleftrightarrow}$

Best response

Optimal for some conjecture about opponents' actions.

Strict dominance

Opponents' profiles: $a_{-i} \in A_{-i} = \prod_{j \neq i} A_j$.

A mixture $s_i \in \Delta(A_i)$ **strictly dominates** an action $a_i \in A_i$ if

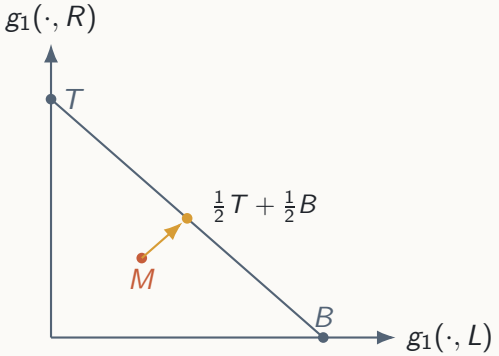
$$g_i(s_i, a_{-i}) > g_i(a_i, a_{-i})$$
for every $a_{-i} \in A_{-i}$.

An action is **strictly dominated** if such a mixture exists.

A mixture can dominate a pure action

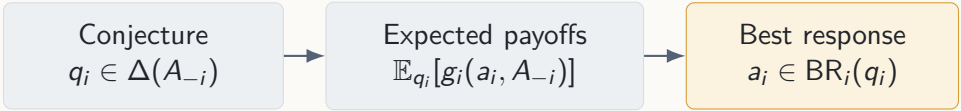
	L	R
T	0	3
M	1	1
B	3	0

Only Player 1's payoffs.



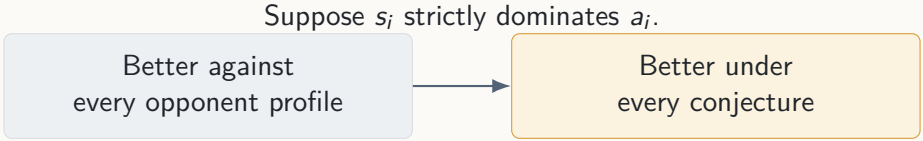
$\frac{1}{2}T + \frac{1}{2}B$ yields $\frac{3}{2} > 1$ against both columns.

Best responses: rational choice needs a conjecture



$$\mathrm{BR}_i(q_i) = \arg\max_{a_i \in A_i} \sum_{a_{-i}} q_i(a_{-i}) g_i(a_i, a_{-i}).$$

Strict dominance rules out a best response



$$\mathbb{E}_{q_i}[g_i(s_i, A_{-i})] > \mathbb{E}_{q_i}[g_i(a_i, A_{-i})] \quad \text{for every } q_i.$$

Dominated iff never a best response

Strictly dominated by a mixture $\iff$ never a best response.

Never a best response

$$\forall q_i \quad \exists b_i :$$
$$\mathbb{E}_{q_i}[g_i(b_i, A_{-i})]$$
$$> \mathbb{E}_{q_i}[g_i(a_i, A_{-i})].$$

One dominating mixture

$$\exists s_i \quad \forall a_{-i} :$$
$$g_i(s_i, a_{-i})$$
$$> g_i(a_i, a_{-i}).$$

The minimax theorem turns local improvements into one uniform improvement.

Proof: build a game of payoff improvements

Fix the action a_i.

$$\gamma(b_i, a_{-i}) = g_i(b_i, a_{-i}) - g_i(a_i, a_{-i}).$$



If a_i is never a best response,

$$\max_{b_i} \mathbb{E}_{q_i}[\gamma(b_i, A_{-i})] > 0 \quad \text{for every } q_i.$$

Continuity and compactness give a strictly positive minimum.

Proof: apply the minimax theorem

$$\underbrace{\min_{q_i} \max_{b_i} \mathbb{E}_{q_i}[\gamma(b_i, A_{-i})]}_{\text{never a best response}} = \underbrace{\max_{s_i} \min_{a_{-i}} \gamma(s_i, a_{-i})}_{\text{one mixture works everywhere}} > 0.$$

Choose a maximizing s_i. Then

$$g_i(s_i, a_{-i}) - g_i(a_i, a_{-i}) > 0 \qquad \text{for every } a_{-i}.$$

s_i strictly dominates a_i.

II. Iterating rationality

I am rational; I know you are rational; I know you know...

Iterated deletion

Delete strictly dominated actions.

Repeat in the reduced game.

Largest fixed point

Actions justify one another.

Take the largest such family.

Do the two approaches select the same actions?

Iterated deletion: two rounds of reasoning

	L	R
T	$3, 2$	$2, 1$
B	$1, 0$	$0, 3$

Rationality
T strictly dominates B
Player 1 chooses T

▼

Knowledge of rationality
Player 2 expects T
Player 2 chooses L

The only surviving profile is (T, L).

Iterated deletion: keep undominated actions

$$R_i^0 = A_i, \qquad R_{-i}^n = \prod_{j \neq i} R_j^n,$$

$$R_i^{n+1} = \left\{ a_i \in R_i^n : a_i \text{ is undominated against } R_{-i}^n \right\}.$$

$$R_i = \bigcap_{n \geq 0} R_i^n \qquad \text{(surviving actions)}.$$

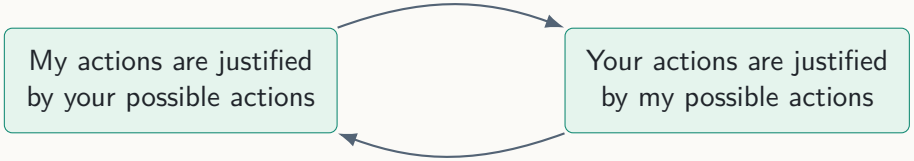


Finite action sets: the decreasing sequence eventually stabilizes.

Largest fixed point: mutual justification

Let $B_{-i} = \prod_{j \neq i} B_j$. A family $(B_i)_i$ is jointly rationalizable if

$$\forall i,\ \forall b_i \in B_i, \quad \exists q_i \in \Delta(B_{-i}) :\ b_i \in \mathrm{BR}_i(q_i).$$



Largest family: $\widetilde{R}_i = \bigcup_{(B_j) \text{ jointly rationalizable}} B_i.$

Fixed point: taking best responses adds no new actions.

Iterated deletion and the largest fixed point coincide

Iterated deletion
Surviving actions
R_i

$\Longleftrightarrow$

Largest fixed point
Rationalizable actions
$\widetilde{R}_i$

$$R_i = \widetilde{R}_i \qquad \text{for every player } i.$$

Proof: the two inclusions

Survivors are justified

Stop: $R_i^{n+1} = R_i^n = R_i$.

Part I: each survivor is justified by a conjecture on R_{-i}.

Maximality: $R_i \subseteq \widetilde{R}_i$.

Justified actions survive

Start: $\widetilde{R}_i \subseteq R_i^0$.

If $\widetilde{R}_j \subseteq R_j^n$ for all j:
conjectures on $\widetilde{R}_{-i}$ are on R_{-i}^n.
Part I gives $\widetilde{R}_i \subseteq R_i^{n+1}$.

Induction: $\widetilde{R}_i \subseteq \bigcap_n R_i^n = R_i$.

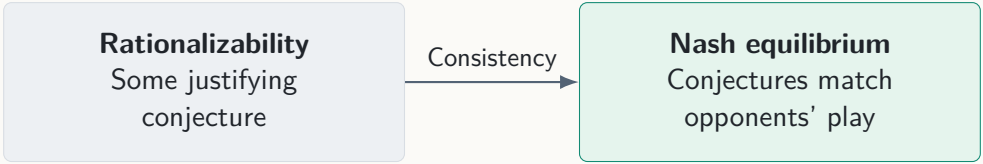
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Signals without payoff news; obedience; revelation

Nash equilibrium: consistent conjectures



$$g_i(s_i, s_{-i}) \geq g_i(s_i', s_{-i}) \qquad \text{for every } i,\ s_i' \in \Delta(A_i).$$

Bayesian games: information and payoffs

Information

States $\omega \in \Omega$.

Common prior $p \in \Delta(\Omega)$.

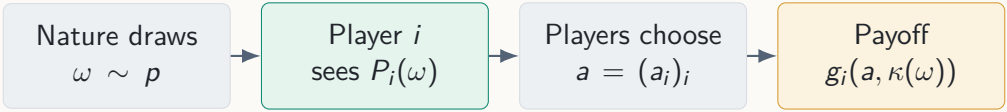
Player i observes the cell $P_i(\omega)$ of a partition $\mathcal{P}_i$.

Payoffs

Payoff state $k = \kappa(\omega)$.

Actions $a_i \in A_i$.

Payoffs $g_i(a, \kappa(\omega))$.



Types are information cells

$$T_i = \mathcal{P}_i \qquad t_i(\omega) = P_i(\omega).$$

$t_i^a = \{\omega_1, \omega_2\}$



Same information: same type

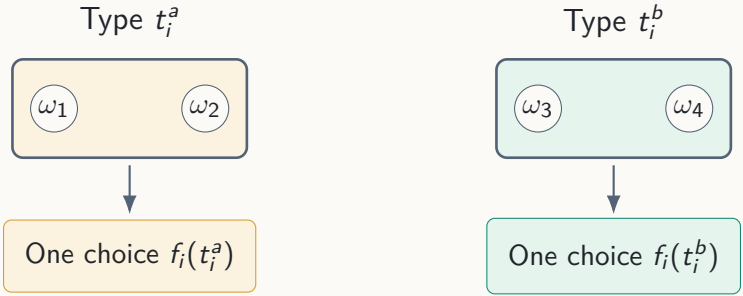
$t_i^b = \{\omega_3, \omega_4\}$



Same information: same type

$$p(t_i) = \sum_{\omega \in t_i} p(\omega).$$

A strategy is a contingent plan



Pure strategy: $f_i : T_i \to A_i$.

Behavioral strategy: $\sigma_i : T_i \to \Delta(A_i)$.

Best response: one type at a time

Bayes's rule: $\omega \in t_i$, $p(t_i) > 0$.

$$p(\omega \mid t_i) = \frac{p(\omega)}{\sum_{\omega' \in t_i} p(\omega')}.$$

Ex ante
Best response among contingent plans

$\Longleftrightarrow$

At every type
Best response whenever $p(t_i) > 0$

Why best responses separate by type

Fix opponents' contingent plans.

$$\underbrace{\mathbb{E}_p[g_i]}_{\text{ex ante}} = \sum_{\substack{t_i \in T_i \\ p(t_i)>0}} p(t_i)\, \underbrace{\mathbb{E}_p[g_i \mid t_i]}_{\text{at type } t_i}.$$

Profitable at one type

Change the action only at that type.

Its positive weight makes the ex ante payoff increase.

Optimal at every type

Each type has its own choice.

No plan can improve the sum.

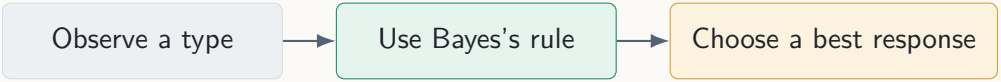
Actions at zero-probability types do not affect expected payoffs.

Bayesian Nash equilibrium

Nash equilibrium in contingent plans:

for every player i and every type t_i with positive probability,

$$\sigma_i(t_i) \in \arg\max_{s_i \in \Delta(A_i)} \mathbb{E}_p\left[g_i\left(s_i, \sigma_{-i}(t_{-i}(\omega)), \kappa(\omega)\right) \mid \omega \in t_i\right].$$



With finite types and actions: Nash's theorem gives existence.

Do you really want to defect?

State L

	C	D
C	$0, -2$	$-10, -1$
D	$-1, -10$	$-5, -5$

State R

	C	D
C	$0, -2$	$-10, -7$
D	$-1, -10$	$-5, -11$

$$\Pr(L) = p, \quad \Pr(R) = 1 - p, \qquad 0 < p < 1.$$

Player 1 chooses the row; Player 2 chooses the column.

Case 1: nobody observes the state

	C	D
C	$0, -2$	$-10, 6p - 7$
D	$-1, -10$	$-5, 6p - 11$

Pure equilibrium (C, C)

$$p \leq \tfrac{5}{6}.$$

Pure equilibrium (D, D)

$$p \geq \tfrac{1}{6}.$$

Two pure equilibria coexist for $\frac{1}{6} \leq p \leq \frac{5}{6}$.

Case 2: only Player 2 observes the state

A plan specifies an action in L, then an action in R.

CC	CD	DC	DD

In state L: choose D

Against C: $-1 > -2$.

Against D: $-5 > -10$.

In state R: choose C

Against C: $-2 > -7$.

Against D: $-10 > -11$.

DC is optimal at each type, whatever Player 1 does.

Player 1 responds to the informed plan

Against DC:

$$\mathbb{E}[g_1(C, DC)] = -10p, \qquad \mathbb{E}[g_1(D, DC)] = -4p - 1.$$

$$C \text{ is optimal} \iff p \leq \tfrac{1}{6}.$$



At $p = \frac{1}{6}$, Player 1 may use any mixture of C and D.

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Signals without payoff information

Fixed payoffs

One game: $g_i(a)$.

No uncertain payoff state.

Private information

Joint signals $(X_i)_i$.

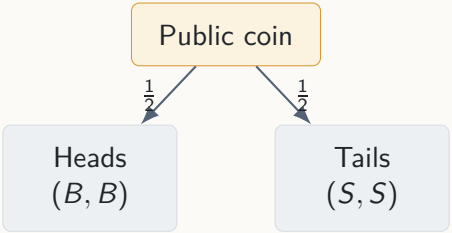
Player i observes only X_i.

Can signals that reveal nothing about payoffs still affect play?

A public signal coordinates play

	B	S
B	$2, 1$	$0, 0$
S	$0, 0$	$1, 2$

Battle of the sexes.



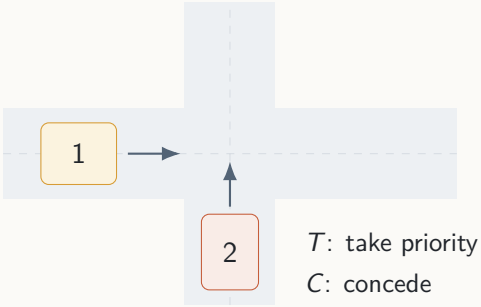
Following the coin is a Bayesian Nash equilibrium.

Both get $\frac{3}{2}$ versus $\frac{2}{3}$ in the mixed Nash equilibrium without signals.

Can private signals do more?

	C	T
C	$6, 6$	$2, 7$
T	$7, 2$	$0, 0$

Chicken: concede or take priority.

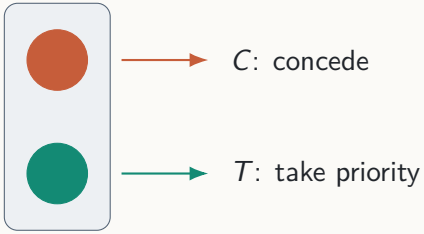


Pure Nash equilibria: (T, C) and (C, T).

Each driver sees only their own traffic signal

	R	G
R	$\frac{1}{3}$	$\frac{1}{3}$
G	$\frac{1}{3}$	0

Joint signal distribution.



A recommendation changes the conditional belief about the other driver.

Best responses at each signal

Green: take priority

Other signal: red.

$$\mathbb{E}[T \mid G] = 7,$$
$$\mathbb{E}[C \mid G] = 6.$$

Red: concede

Other signal: red or green, each with probability $\frac{1}{2}$.

$$\mathbb{E}[C \mid R] = \tfrac{1}{2}(6+2) = 4,$$
$$\mathbb{E}[T \mid R] = \tfrac{1}{2}(7+0) = \tfrac{7}{2}.$$

These plans form a Bayesian Nash equilibrium of the signal game.

Private recommendations improve both payoffs

$$\mu(C,C) = \mu(C,T) = \mu(T,C) = \tfrac{1}{3}, \qquad \mu(T,T) = 0.$$

	Player 1	Player 2
Fair lottery over the pure Nash equilibria	$\frac{9}{2}$	$\frac{9}{2}$
Symmetric mixed Nash equilibrium	$\frac{14}{3}$	$\frac{14}{3}$
Private traffic signals	5	5

$$\mathbb{E}[g_i] = \tfrac{1}{3}(6+2+7) = 5.$$

Obedience: correlated equilibrium

Aumann (1974)

A mediator draws $a \sim \mu \in \Delta(A)$. Player i's type is the recommendation a_i.

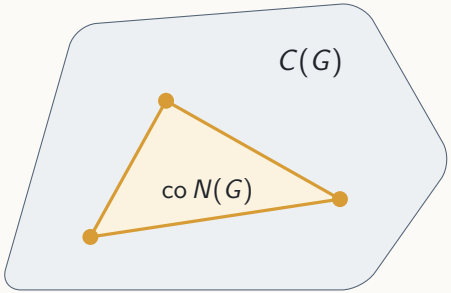
$$\sum_{a_{-i}} \mu(a_i, a_{-i})\big[g_i(a_i, a_{-i}) - g_i(b_i, a_{-i})\big] \geq 0$$

for every i, a_i, b_i.

If $\Pr(a_i) > 0$:

$$\mathbb{E}[g_i(a_i, A_{-i}) \mid a_i] \geq \mathbb{E}[g_i(b_i, A_{-i}) \mid a_i].$$

A convex set of obedient distributions



Schematic

$N(G)$: Nash outcome distributions.
$C(G)$: correlated-equilibrium distributions.

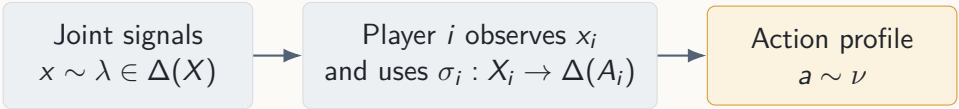
Obedience inequalities are linear in μ.

$C(G)$ is a compact convex polytope.

Nash distributions satisfy obedience; convexity also allows lotteries over them.

$$N(G) \subseteq \text{co}\, N(G) \subseteq C(G).$$

A Bayesian equilibrium with arbitrary signals



$$\nu(a) = \sum_x \lambda(x) \prod_i \sigma_i(a_i \mid x_i).$$

Fixed payoffs; $\sigma = (\sigma_i)_i$ is a Bayesian Nash equilibrium.

The revelation principle for correlated equilibrium

Equilibrium with arbitrary private signals

$\Longleftrightarrow$ an obedient distribution of action recommendations.

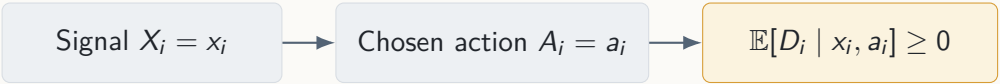


The action distribution ν is preserved.

Proof: keep the signal while checking a deviation

Fix a player i, a recommended action a_i, and a deviation b_i.

$$D_i = g_i(a_i, A_{-i}) - g_i(b_i, A_{-i}).$$



Every action used at a signal is a best response.
Own randomization carries no extra information about opponents.

Proof: average over the signals behind a recommendation

$$\mathbb{E}[D_i \mid A_i = a_i] = \sum_{x_i} \Pr(x_i \mid A_i = a_i)\, \underbrace{\mathbb{E}[D_i \mid X_i = x_i, A_i = a_i]}_{\geq 0} \geq 0.$$

Best response at each underlying signal → Average → Best response given the recommendation alone

$$\nu \in C(G).$$

Conversely: draw recommendations from ν; obedience gives best responses.

Lecture 1: averaging conditional expectations.

Main references

Minimax

von Neumann (1928).

Chiappori, Levitt and Groseclose (2002).

Rationalizability

Bernheim (1984); Pearce (1984).

Rationalizable strategic behavior.

Bayesian games

Nash (1951), *Non-Cooperative Games*.

Harsanyi (1967–68): games with incomplete information.

Correlated equilibrium

Aumann (1974), *Subjectivity and Correlation in Randomized Strategies*.

Maschler, Solan and Zamir (2013), *Game Theory*.

Conclusion

Minimax	Guarantee equals defense
Rationalizability	Common knowledge of rationality
Bayesian games	Best responses, type by type
Correlated equilibrium	Payoff-irrelevant signals can change play

Lecture 3: how do we compare information, value it, and choose it?

Weak dominance: the order can matter

	L	R
T	$1, 0$	$0, 2$
M	$0, 2$	$2, 0$
B	$1, 1$	$2, 1$

Delete T first

Then L weakly dominates R.

Delete R, then M: (B, L).

Delete M first

Then R weakly dominates L.

Delete L, then T: (B, R).

Conjectures: must opponents' actions be independent?

Player 1 chooses T, M, B; Player 2 chooses L, R; Player 3 chooses l, r.

Player 3 chooses l

	L	R
T	-8	0
M	0	0
B	-3	-3

Player 3 chooses r

	L	R
T	0	0
M	0	-8
B	-3	-3

Only Player 1's payoffs are displayed.

The same marginals can justify different choices

Independent conjecture

$p = \Pr(L)$, $q = \Pr(l)$.

$$\mathbb{E}[T] = -8pq,$$
$$\mathbb{E}[M] = -8(1-p)(1-q).$$

$$\max\{\mathbb{E}[T], \mathbb{E}[M]\} \geq -2 > -3.$$

B is never a best response.

Correlated conjecture

	l	r
L	$\frac{1}{2}$	0
R	0	$\frac{1}{2}$

$$\mathbb{E}[T] = \mathbb{E}[M] = -4.$$

B is the unique best response.

At $p = q = \frac{1}{2}$, both conjectures have uniform marginals.

The bound for independent conjectures

Cauchy–Schwarz:

$$\sqrt{pq} + \sqrt{(1-p)(1-q)} \le \sqrt{p + (1-p)}\,\sqrt{q + (1-q)} = 1.$$

Hence

$$\min\{\sqrt{pq}, \sqrt{(1-p)(1-q)}\} \le \tfrac{1}{2}.$$

$$\min\{pq, (1-p)(1-q)\} \le \tfrac{1}{4}.$$

The full game of contingent plans

Only Player 2 is informed; first action in L, second action in R.

	CC	CD	DC	DD
C	$0, -2$	$10p - 10, 5p - 7$	$-10p, p - 2$	$-10, 6p - 7$
D	$-1, -10$	$4p - 5, p - 11$	$-4p - 1, 5p - 10$	$-5, 6p - 11$

DC is strictly dominant when $0 < p < 1$.

At $p = 0$ or $p = 1$, the plan at the zero-probability type is irrelevant.

Exercise: who knows the quality of the car?

Car values

	Seller v_s	Buyer v_b
Good car	3	4
Bad car	0	0

Each quality has probability $\frac{1}{2}$.

Trade at price π

Seller: keep or sell. Buyer: decline or buy.

Trade requires both to agree.

Net gains: $(\pi - v_s,\ v_b - \pi)$.

No trade: payoff 0 for each.

Compare: both informed / neither informed / only seller informed.

For which prices can trade occur in equilibrium?

Why the maxmin never exceeds the minmax

For any s_1, s_2,

$$\min_{t_2} g(s_1, t_2) \leq g(s_1, s_2) \leq \max_{t_1} g(t_1, s_2).$$

Thus, for every s_2,

$$\max_{s_1} \min_{t_2} g(s_1, t_2) \leq \max_{t_1} g(t_1, s_2).$$

$$\max_{s_1} \min_{s_2} g(s_1, s_2) \leq \min_{s_2} \max_{s_1} g(s_1, s_2).$$